

Review of Last Lecture

4.7 System Function

4.8 Time-domain Characteristics from System Function Zero-pole Distribution

4.9 Frequency Response from System Function Zero-pole Distribution

4.7 System Function



$H(s) = \int_{0^-}^{\infty} h(t)e^{-st} dt$ is the Laplace transform of the unit impulse response of a **causal system**, also known as the **system function**.

For a continuous-time signal $x(t)$, acting on a linear time-invariant system with unit impulse response $h(t)$, its **zero-state response** $y_{zs}(t)$ is equal to the **convolution** of the input signal and the unit impulse response:

$$\begin{array}{c} x(t) \longrightarrow \boxed{h(t)} \longrightarrow y_{zs}(t) = h(t) * x(t) \end{array}$$

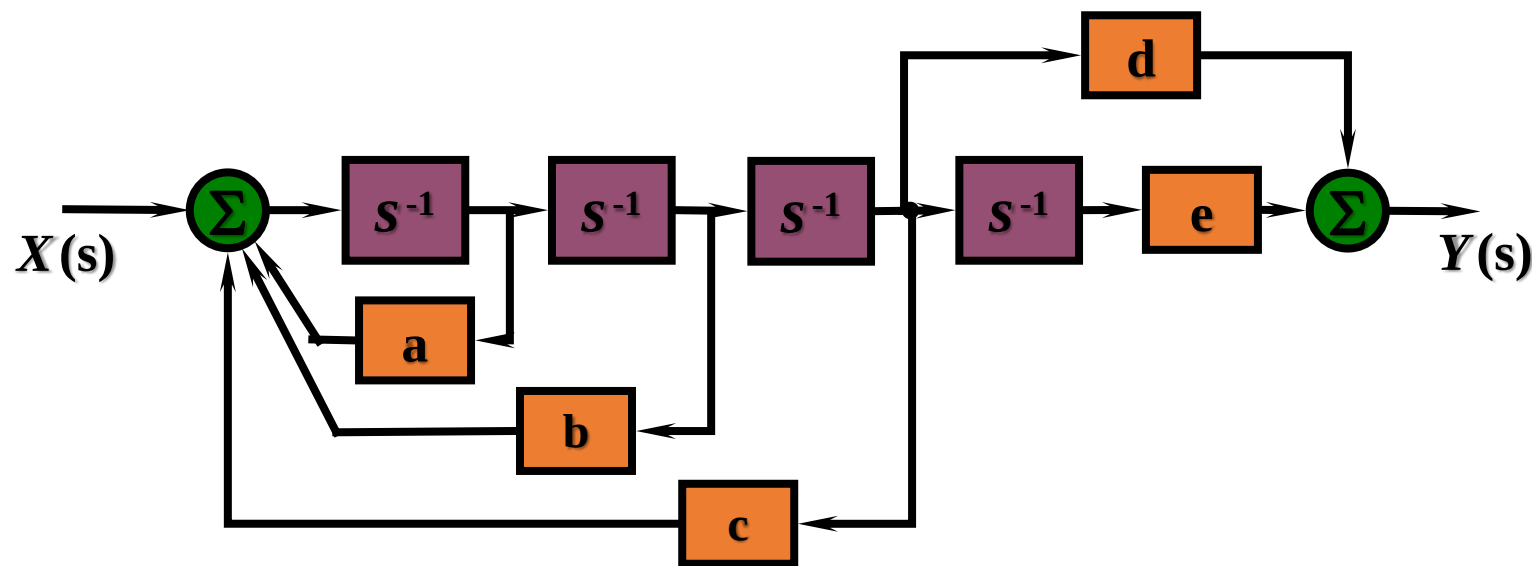
By the **time-domain convolution theorem**, the Laplace transform of the zero-state response is equal to the **product** of the Laplace transform of the input signal and the Laplace transform of the unit impulse response:

$$\begin{array}{c} X(s) \longrightarrow \boxed{H(s)} \longrightarrow Y_{zs}(s) = H(s)X(s) \end{array}$$

The **system function** is equal to the **ratio** of the Laplace transform of the zero-state response to the Laplace transform of the input signal:

$$H(s) = Y_{zs}(s) / X(s)$$

Given $H(s) = \frac{2s+3}{s(s+3)(s+2)^2}$, find the parameters in the s-domain block diagram:
 $a =$ [填空1], $b =$ [填空2], $c =$ [填空3], $d =$ [填空4], $e =$ [填空5].



作答

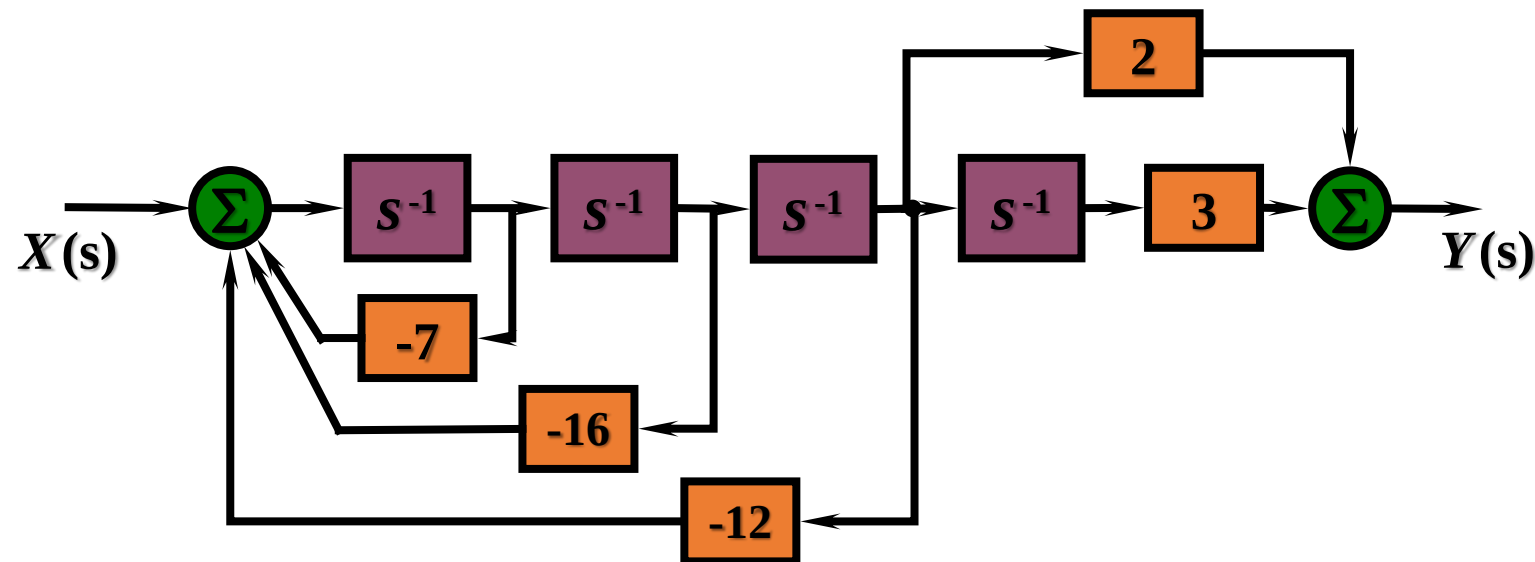
4.7 System Function



Given $H(s) = \frac{2s+3}{s(s+3)(s+2)^2}$, find the parameters in the s-domain block diagram.

Solution: The system function can be transformed into:

$$H(s) = \frac{2s+3}{s(s+3)(s+2)^2} = \frac{2s+3}{s^4 + 7s^3 + 16s^2 + 12s} = \frac{2s^{-3} + 3s^{-4}}{1 + 7s^{-1} + 16s^{-2} + 12s^{-3}}$$

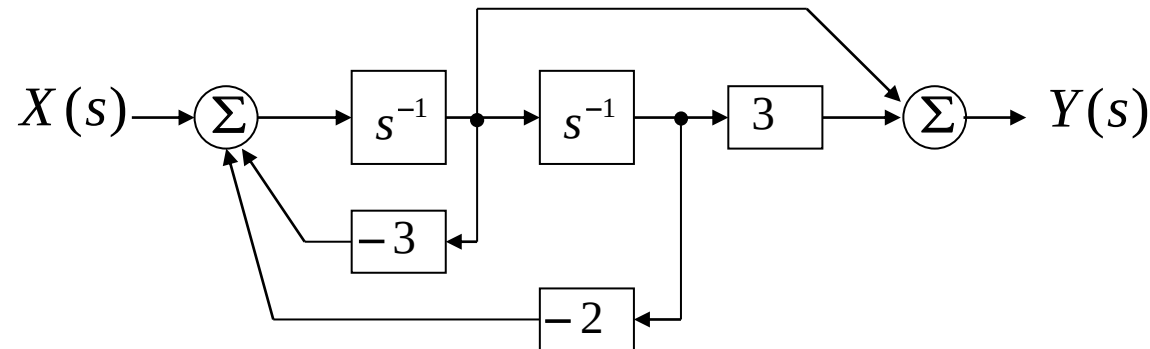


Roles of System Function

1. To find the **zero-state response** of the system: $Y_{zs}(s) = H(s)X(s)$ $y_{zs}(t) = \mathcal{L}T^{-1}[Y_{zs}(s)]$
2. To relate the system function to the system's **differential equation** and **simulation block diagram**.

$$H(s) = \frac{s^{-1} + 3s^{-2}}{1 + 3s^{-1} + 2s^{-2}} = \frac{s + 3}{s^2 + 3s + 2}$$

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} + 3x(t)$$



3. To determine **the impulse response** and characteristics of **response components** by the zero-pole distribution of the system function.
4. To determine the system's **frequency response characteristics** by the zero-pole distribution of the system function.

Zeros, Poles of System Function & Excitation, and Response Components

The **poles** of the system function and excitation jointly determine the response waveform:

$$Y(s) = X(s)H(s)$$

Poles of the system function correspond to free response components; poles of the excitation correspond to forced response components.

If **zeros and poles of the system function and excitation cancel each other**, the corresponding time component does not appear in the response.

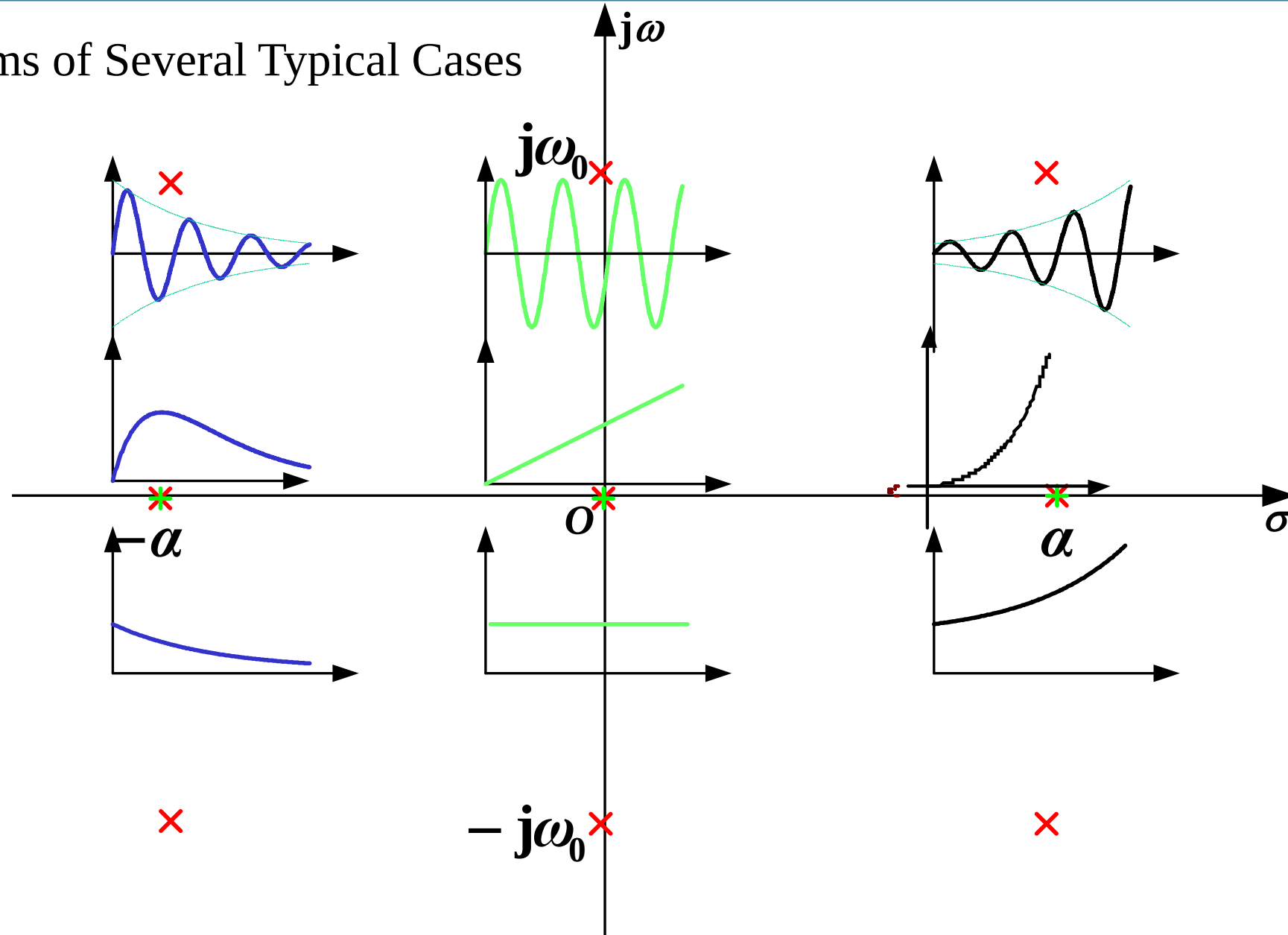
Example: Laplace transform of zero-state response

$$Y(s) = X(s)H(s) = \frac{1}{s+3} \cdot \frac{s+3}{s^2+3s+2}$$

$$\text{zero-state response: } y(t) = (e^{-t} - e^{-2t})u(t)$$

Only free response exists, as forced response is zero due to **zero-pole cancellation**.

Diagrams of Several Typical Cases





Correspondence between Poles of $H(s)$ and Waveform Characteristics of $h(t)$:

Poles on the left half-plane of the s-plane

$h(t)$ exhibits a decaying form

Poles in the right half-plane of the s-plane

$h(t)$ exhibits a growing form

Poles on the imaginary axis of the s-plane
(excluding the origin)

$h(t)$ has constant-amplitude oscillation
(first-order poles) or growing
oscillation (multi-order poles)

Poles on the real axis of the s-plane
(excluding the origin)

$h(t)$ exhibits exponential-related
changes (first-order or multi-
order poles)

Poles at the origin of the s-plane

$h(t)$ is equal to $u(t)$ (first-order pole)
or grows as t^n (multi-order poles)

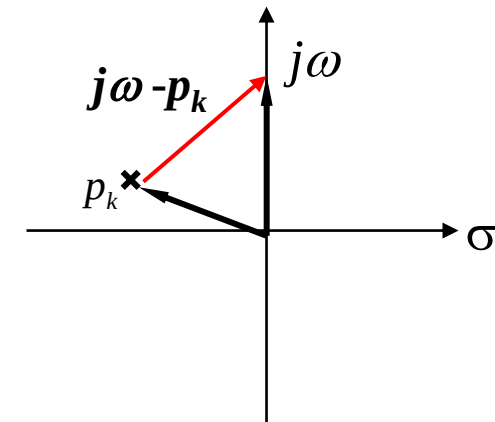
4.9 Frequency Response from System Function Zero-pole Distribution

The frequency response of a system can **generally** be expressed:

$$H(\omega) = H(s) \Big|_{s=j\omega} = \frac{G \prod_{k=1}^M (s - z_k)}{\prod_{k=1}^N (s - p_k)} \Bigg|_{s=j\omega} = \frac{G \prod_{k=1}^M (j\omega - z_k)}{\prod_{k=1}^N (j\omega - p_k)} = \frac{G \prod_{k=1}^M B_k}{\prod_{k=1}^N A_k} e^{j(\sum_{k=1}^M \beta_k - \sum_{k=1}^N \theta_k)} = |H(\omega)| e^{j\varphi(\omega)}$$

$$|H(\omega)| = \frac{G \prod_{k=1}^M B_k}{\prod_{k=1}^N A_k}$$

$$\varphi(\omega) = \sum_{k=1}^M \beta_k - \sum_{k=1}^N \theta_k$$



The system's magnitude-frequency response is **the ratio of the product of the magnitudes of zero vectors to that of pole vectors**.

The system's phase-frequency response is **the sum of the phase angles of zero vectors minus that of pole vectors**.

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4.10 Zero-Pole Locations of All-pass and Minimum-phase-shift Functions

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4.12 Bilateral Laplace Transform

4.13 Relationship between Laplace Transform and Fourier Transform

Learning Objectives of This Lecture

1. Understand the characteristics of zero and pole distribution of **all-pass** functions and **minimum-phase-shift** functions;
2. Skillfully use **system functions** to determine the **stability** of systems;
3. Be familiar with the correspondence between the **region of convergence** of bilateral Laplace transform and the **original function**;
4. Understand the correspondence between Laplace transform and **Fourier transform**.

4.10.1 Zero-Pole Distribution of All-pass Systems

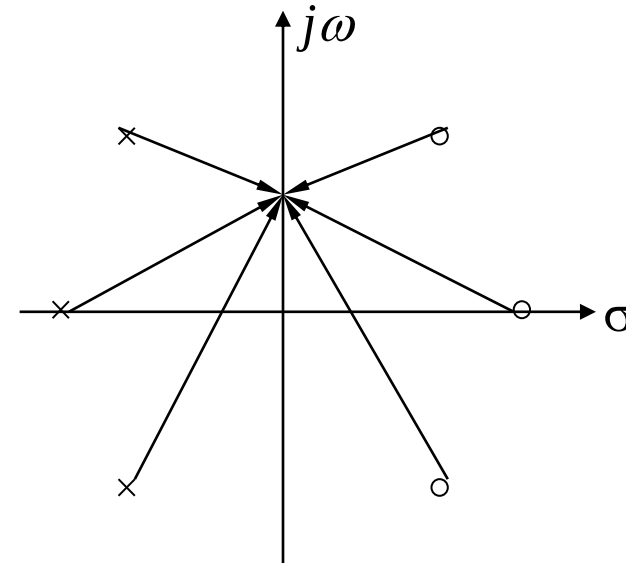
The **magnitude-frequency response** of an all-pass system is a **constant** at all frequencies.

The **phase-frequency response** of an all-pass system is **not restricted**.

The product of the magnitudes of the zero vectors and the product of the magnitudes of the pole vectors of an **all-pass system** function are equal **at all frequencies**.

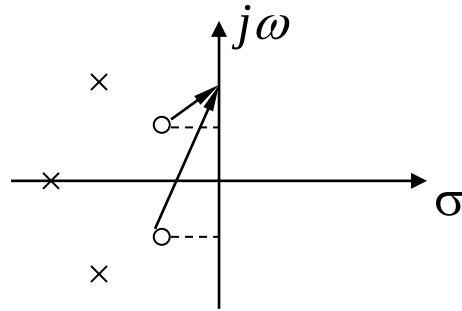
Hence, **the zeros and poles are symmetric with respect to the imaginary axis**.

$$|H(\omega)| = \frac{G \prod_{k=1}^N B_k}{\prod_{k=1}^N A_k} = G$$

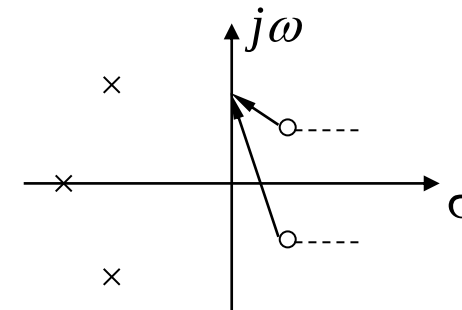


4.10.2 Zero-Pole Distribution of Minimum-phase-shift Systems

Minimum-phase-shift System: All zeros of the system function are located on the left half-plane or on the imaginary axis of the s-plane. Otherwise it is a non-minimum-phase-shift system.



Pole-Zero Plot of
Minimum-phase-shift System



Pole-Zero Plot of
Non-minimum-phase-shift System

In the above two pole-zero plots, poles are distributed the same, zeros have equal imaginary parts but opposite-signed real parts. Obviously, **their magnitude-frequency responses are identical**; at all frequencies, the phase angles of zeros in the left plot are smaller than those in the right plot, while pole phase angles are the same. Thus, **the left plot has a smaller phase shift** in absolute value.

Among the following system functions, which one belongs to a minimum-phase-shift system? ()

A $H(s) = \frac{(s-2)^2+6}{(s+1)(s+5)^2}$

B $H(s) = \frac{(s-1)(s+2)}{s(s+1)^2}$

C $H(s) = \frac{(s+2)^2+6}{(s+1)(s+5)^2}$

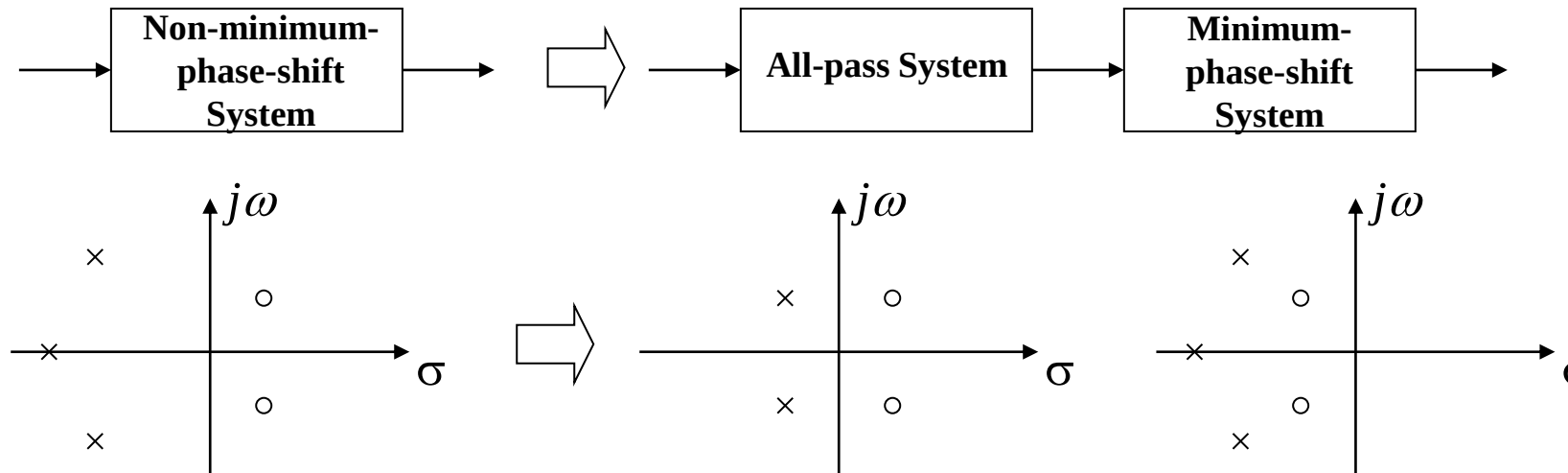
D $H(s) = \frac{(s-3)^2}{s^3+5s+7}$

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4.10 Zero-Pole Locations of All-pass and Minimum-phase-shift Functions

Any non-minimum-phase-shift system can be expressed as **a cascade of an all-pass system and a minimum-phase-shift system.**

$$H(s) = H_{AP}(s) \cdot H_{\min}(s)$$



The **poles** of the all-pass system **cancel** with the **zeros** of the minimum-phase-shift system.

Example:

$$H(s) = \frac{(s - \alpha_2)^2 + \Omega_2^2}{(s + \alpha_0)[(s + \alpha_1)^2 + \Omega_1^2]} = \frac{(s + \alpha_2)^2 + \Omega_2^2}{(s + \alpha_0)[(s + \alpha_1)^2 + \Omega_1^2]} \cdot \frac{(s - \alpha_2)^2 + \Omega_2^2}{(s + \alpha_2)^2 + \Omega_2^2}$$

$$H_{AP}(s) = \frac{(s - \alpha_2)^2 + \Omega_2^2}{(s + \alpha_2)^2 + \Omega_2^2}$$

$$H_{\min}(s) = \frac{(s + \alpha_2)^2 + \Omega_2^2}{(s + \alpha_0)[(s + \alpha_1)^2 + \Omega_1^2]}$$

4.11 Stability of Linear Systems

4.11.1 Causality and Stability of Systems

The **stability** of a system refers to a system where **a bounded input** produces **a bounded output**. That is, if the input $|x(\cdot)| < \infty$, the output $|y(\cdot)| < \infty$, the system must be stable. Specifically, **the output cannot contain impulse functions and their derivatives**.

The stability of a linear time-invariant (LTI) system means its impulse response satisfies absolute integrability, i.e.

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty$$

The **causality** of a system means that when the initial state is zero, the output does not occur before the input acts on the system. That is, if $t < t_0$ and $x(t)=0$, then necessarily $t < t_0$ and $y(t)=0$.

The causality of a linear system means its impulse response satisfies:

$$h(t) = h(t)u(t)$$

4.11 Stability of Linear Systems

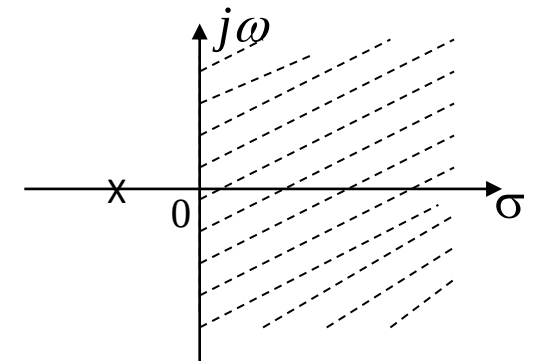
When a system is **causal and stable**, its unit impulse response should satisfy absolute integrability from 0^- to ∞ :

$$\int_{0^-}^{\infty} |h(t)| dt < \infty \quad (1)$$

$$\text{As } H(s) = \int_{0^-}^{\infty} h(t) e^{-st} dt = \int_{0^-}^{\infty} h(t) e^{-(\sigma + j\omega)t} dt \quad (2)$$

$$\text{For (2) to be absolutely integrable,} \quad \int_{0^-}^{\infty} |h(t) e^{-\sigma t}| dt < \infty \quad (3)$$

$$\left(|e^{-j\omega t}| = 1 \right)$$



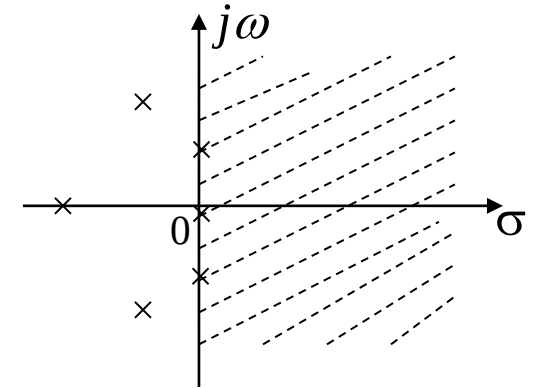
By comparing (1) and (3), it can be seen that the minimal region of convergence (ROC) of the system function of a linear causal and stable system is $\sigma = \text{Re}\{s\} > 0$.

That is, **the poles of the system function of a causal and stable system can only be distributed on the left half-plane of the imaginary axis on the s-plane.**

4.11 Stability of Linear Systems

What if the poles of the system function are located on the imaginary axis of the s-plane?

- **Case 1: a first-order pole at the origin** corresponds to the factor $\frac{1}{s}$, whose inverse transform is the unit step signal $u(t)$, which does not satisfy absolute integrability. However, **when $t > 0$, its amplitude remains stable.**
- **Case 2: a pair of complex conjugate poles on the imaginary axis** correspond to factors like $\frac{\Omega_0}{s^2 + \Omega_0^2}$ or $\frac{s}{s^2 + \Omega_0^2}$, whose inverse Laplace transforms are $\sin(\Omega_0 t)$ and $\cos(\Omega_0 t)$, respectively, not satisfying absolute integrability. But **when $t > 0$, their maximum amplitude remains stable.**
- **Case 3:** If the poles on the imaginary axis are of multiple orders, the corresponding time signals does not satisfy absolute integrability, and their amplitude increases when $t > 0$.



Thus, **the poles on the imaginary axis make the system unstable.** The cases of **first-order pole and complex conjugate poles on the imaginary axis** are also referred to as **marginally stable.**

4.11 Stability of Linear Systems

A stable system also has requirements for **the number of zeros** in its system function.

Let the system function be:

$$H(s) = \frac{N(s)}{D(s)} = s + \frac{A(s)}{D(s)}$$

$A(s)/D(s)$ is a proper rational fraction. When the bounded input $x(t)=u(t)$ (with its Laplace transform being $1/s$) is applied, the Laplace transform of the output will contain 1, corresponding to the impulse function $\delta(t)$ in the output, which is unbounded in amplitude. It can be seen that the above system is unstable. **$h(t)$ can contain impulse functions but not their derivatives.**

In summary, **the stability conditions in the s-domain for a linear system** are:

- (1) All poles of the system function lie in the left half-plane of the s-plane's imaginary axis; real roots are negative real numbers, and complex roots have negative real parts.
- (2) The order of the system function's numerator polynomial is no higher than that of the denominator (the number of zeros does not exceed the number of poles).

Given a system with the system function $H(s) = \frac{s}{s^2 - s - 6}$, this system is ()

- ☐ A Stable
- ☒ B Unstable

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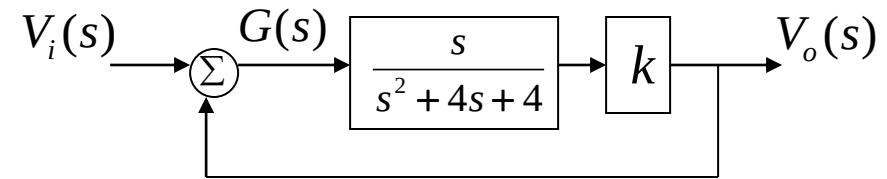
4.11 Stability of Linear Systems

Example 4-30: The following figure shows a feedback system. 1) Find the system function. 2) What condition should k satisfy for the system to be stable? 3) When the system is marginally stable, find the impulse response of the system.

Solution: 1) $G(s) = V_i(s) + V_o(s)$

$$V_o(s) = \frac{ksG(s)}{s^2 + 4s + 4}$$

$$\therefore H(s) = \frac{V_o(s)}{V_i(s)} = \frac{ks}{s^2 + (4-k)s + 4}$$



2) Poles of the system function: $p_{1,2} = \frac{-(4-k) \pm \sqrt{(4-k)^2 - 16}}{2}$

when $k > 4$, the system function has poles with positive real parts, so the system is unstable. Therefore, the system is stable when $k < 4$.

3) when $k = 4$, the system is marginally stable. The system function is $H(s) = \frac{4s}{s^2 + 4}$.

$$\therefore h(t) = 4 \cos 2tu(t)$$

4.11 Stability of Linear Systems

How to determine whether all roots of a polynomial have negative real parts?

When the polynomial is of first order or second order, the following conditions can be used to determine that **all its roots have negative real parts**:

(1) All coefficients of the denominator polynomial are non-zero, i.e., there are no missing terms;

(2) The signs of all coefficients of the denominator polynomial are the same. When the leading coefficient is 1, all other coefficients should be positive.

According to the above conditions, the stability condition of the previous example can be obtained immediately:

$$H(s) = \frac{ks}{s^2 + (4-k)s + 4} \quad 4-k > 0 \quad \therefore k < 4$$

4.11 Stability of Linear Systems

Transient Response and Steady-State Response of Linear Stable Systems

- **Transient Response:** Under the action of excitation (**step signal or causal periodic signal**), it is a component that appears instantaneously in the total response and gradually disappears as time increases.
- **Steady-State Response:** when $t \rightarrow \infty$, it is the component retained in the total response, usually composed of **step functions** and **periodic functions**.
- The poles of the system function of a stable system **lie on the left half-plane of the s-plane**, and the free response function shows an attenuating trend. **Free response = Transient response.**
- If the poles of the excitation **lie on the imaginary axis or in the right half-plane of the s-plane**, **Forced response = Steady-state response.**

Example: Excitation signal $x(t) = \cos(2t)u(t)$, the poles of the system function are -6 and -1, and the total response is

$$r(t) = \underbrace{-\frac{6}{25}e^{-t} + \frac{9}{50}e^{-6t}}_{\substack{\text{Free response} \\ \text{Transient Response}}} + \underbrace{\frac{21}{50}\sin(2t) + \frac{3}{50}\cos(2t)}_{\substack{\text{Forced response} \\ \text{Steady-state response}}} \quad (t \geq 0)$$

4.12 Bilateral Laplace Transform

Bilateral Laplace Transform:

$$F_B(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt = \int_{-\infty}^{\infty} f(t)e^{-\sigma t}e^{-j\omega t}dt$$

For the attenuation factor σ , the situation when $t > 0$ is exactly opposite to that when $t < 0$. Therefore, **the integral result of the bilateral Laplace transform does not necessarily exist**, which is different from the unilateral Laplace transform. We need to discuss the existence of the bilateral Laplace transform.

4.12.1 Region of Convergence (ROC) of Bilateral Laplace Transform

The **ROC** of a signal's Laplace transform is related to the form of the signal itself. According to the definition of the Laplace transform, the existence of the Laplace transform for a general signal should satisfy:

$$\lim_{t \rightarrow \infty} x(t)e^{-\sigma t} = 0 \quad \& \quad \lim_{t \rightarrow -\infty} x(t)e^{-\sigma t} = 0$$

4.12 Bilateral Laplace Transform

4.12.2 Bilateral Laplace Transform of Right-Sided Signals

When a signal is a **right-sided signal**, i.e. $x(t)=0$ for $t < t_0$, (**causal signals** are **right-sided signals**), its Laplace transform is:

$$X(s) = \int_{t_0}^{\infty} x(t)e^{-st} dt = \int_{t_0}^{\infty} x(t)e^{-\sigma t} e^{-j\omega t} dt$$

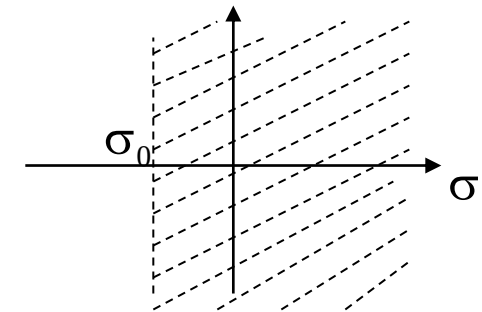
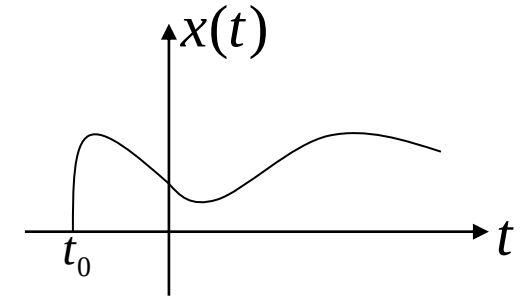
if σ_0 makes the above integral absolutely integrable, i.e., $\int_{t_0}^{\infty} |x(t)e^{-\sigma_0 t} e^{-j\omega t}| dt = \int_{t_0}^{\infty} |x(t)e^{-\sigma_0 t}| dt < \infty$

Let $\sigma = \sigma_0 + \Delta$, given $\lim_{t \rightarrow \infty} x(t)e^{-\sigma_0 t} = 0$, for $x(t)e^{-\sigma t}$ to be absolutely integrable,

we have: $\lim_{t \rightarrow \infty} x(t)e^{-\sigma t} = \lim_{t \rightarrow \infty} x(t)e^{-\sigma_0 t} e^{-\Delta t} = 0$,

which requires $\Delta > 0$, i.e. $\sigma > \sigma_0$.

Thus, the region of convergence (ROC) is $\text{Re}\{s\} = \sigma > \sigma_0$



Given the signal $x(t) = e^{-t}u(t) - e^{-2t}u(t)$, its bilateral Laplace transform and ROC are ()

☐ A $X(s) = \frac{-1}{(s-1)(s-2)}, \sigma > -2$

☒ B $X(s) = \frac{1}{(s+1)(s+2)}, \sigma > -1$

☐ C $X(s) = \frac{-1}{(s-1)(s-2)}, \sigma > -1$

☐ D $X(s) = \frac{1}{(s+1)(s+2)}, \sigma > -2$

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4.12 Bilateral Laplace Transform

Example 4-31: Find the bilateral Laplace transform of the following causal signal and specify its ROC.

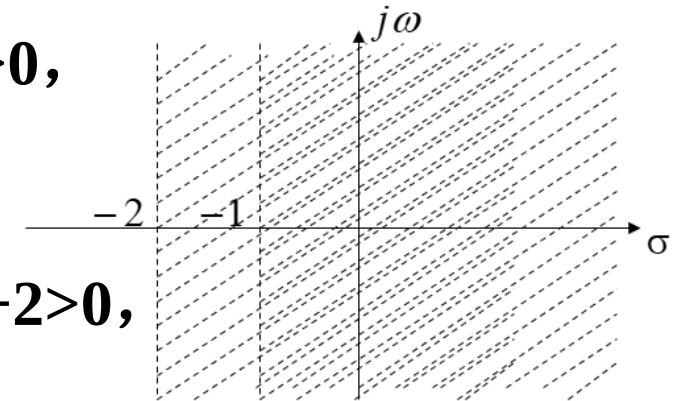
$$x(t) = e^{-t}u(t) - e^{-2t}u(t)$$

Solution:

$$\begin{aligned} X(s) &= \int_{-\infty}^{\infty} x(t)e^{-st} dt = \int_0^{\infty} e^{-t}e^{-st} dt - \int_0^{\infty} e^{-2t}e^{-st} dt \\ &= \int_0^{\infty} e^{-(s+1)t} dt - \int_0^{\infty} e^{-(s+2)t} dt = \int_0^{\infty} e^{-(\sigma+1)t} e^{-j\omega t} dt - \int_0^{\infty} e^{-(\sigma+2)t} e^{-j\omega t} dt \end{aligned}$$

For the first term, the integral is integrable when $\mathbf{Re\{s+1\}=\sigma+1>0}$,
and $\frac{1}{s+1}$ has an ROC of $\mathbf{Re\{s\}=\sigma>-1}$

For the second term, the integral is integrable when $\mathbf{Re\{s+2\}=\sigma+2>0}$,
and $\frac{1}{s+2}$ has an ROC of $\mathbf{Re\{s\}=\sigma>-2}$



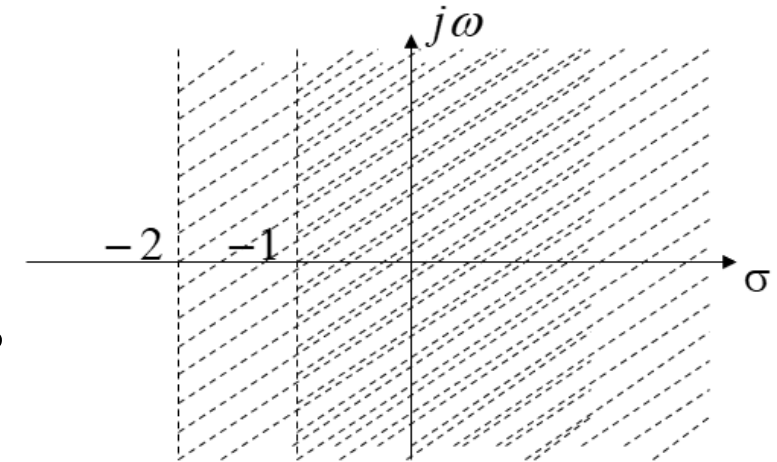
4.12 Bilateral Laplace Transform

Thus, the Laplace transform of the entire function is:

$$X(s) = \frac{1}{s+1} - \frac{1}{s+2} = \frac{1}{(s+1)(s+2)}$$

The ROC is $\text{Re}\{s\} = \sigma > -1$

That is, the ROC of the signal's Laplace transform is **the common region of the ROCs of the two parts.** The ROC of a right-sided signal is to the right of its rightmost pole.



4.12 Bilateral Laplace Transform

4.12.3 Bilateral Laplace Transform of Left-Sided Signals (Anti-Causal Signals)

When a signal is a **left-sided signal**, i.e. $x(t)=0$ for $t>t_0$ (it is an **anti-causal signal** if $t_0=0$), its Laplace transform is:

$$X(s) = \int_{-\infty}^{t_0} x(t)e^{-st} dt = \int_{-\infty}^{t_0} x(t)e^{-\sigma t} e^{-j\omega t} dt$$

if σ_1 makes the above integral absolutely integrable, i.e.,

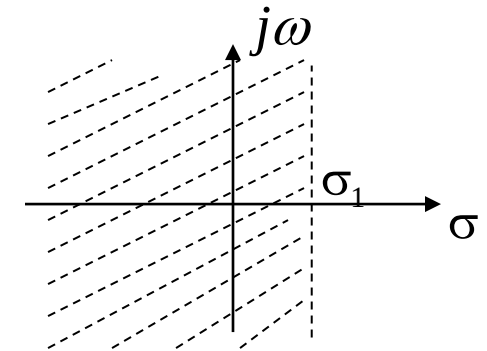
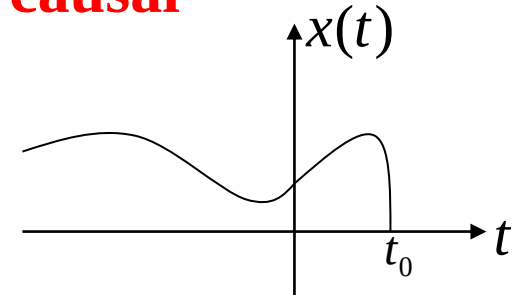
$$X(s) = X(\sigma_1 + j\omega) = \int_{-\infty}^{t_0} x(t)e^{-\sigma_1 t} e^{-j\omega t} dt$$

exists, then the ROC is $\text{Re}\{s\} = \sigma < \sigma_1$

Proof: Let $\sigma = \sigma_1 + \Delta$. Given $\lim_{t \rightarrow -\infty} x(t)e^{-\sigma_1 t} = 0$,

for $x(t)e^{-\sigma t}$ to be absolutely integrable, we have:

$$\lim_{t \rightarrow -\infty} x(t)e^{-\sigma t} = \lim_{t \rightarrow -\infty} x(t)e^{-\sigma_1 t} e^{-\Delta t} = 0, \text{ which requires } \Delta < 0, \text{ i.e. } \sigma < \sigma_1.$$



4.12 Bilateral Laplace Transform

Example 4-32: Given the anti-causal signal $x(t) = -e^{-\alpha t}u(-t)$, find its bilateral Laplace transform $X(s)$.

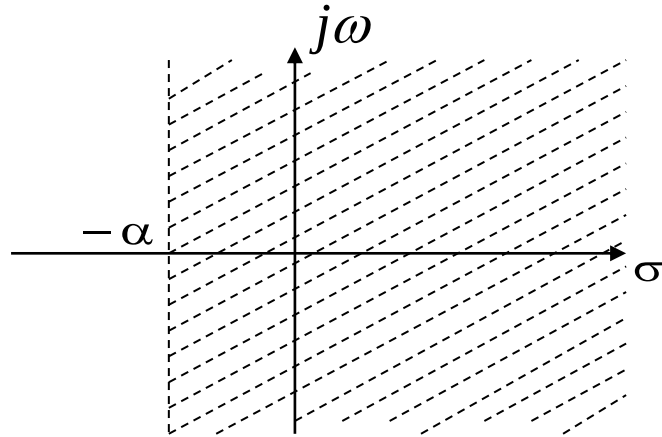
Solution: By definition

$$\begin{aligned} X(s) &= \int_{-\infty}^{\infty} x(t)e^{-st} dt = - \int_{-\infty}^{\infty} e^{-\alpha t} u(-t) e^{-st} dt = - \int_{-\infty}^0 e^{-\alpha t} e^{-st} dt \\ &= - \int_{-\infty}^0 e^{-(s+\alpha)t} dt \end{aligned}$$

when $\text{Re}\{s+\alpha\} = \sigma + \alpha < 0$, $\sigma < -\alpha$, the integral exists.

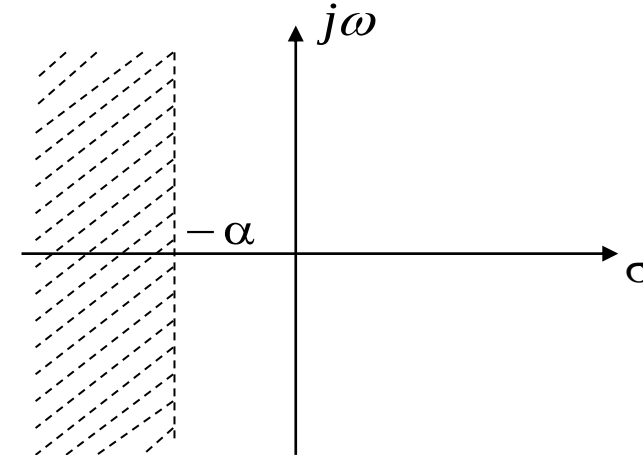
$$X(s) = \frac{1}{(s+\alpha)} e^{-(s+\alpha)t} \Big|_{-\infty}^0 = \frac{1}{(s+\alpha)}, \quad \sigma < -\alpha$$

4.12 Bilateral Laplace Transform



Bilateral Laplace Transform of $e^{-\alpha t}u(t)$

$$e^{-\alpha t}u(t) \xleftrightarrow{LT} 1/(s + \alpha), \quad \text{Re}\{s\} > -\alpha$$



Bilateral Laplace Transform of $-e^{-\alpha t}u(-t)$

$$-e^{-\alpha t}u(-t) \xleftrightarrow{LT} 1/(s + \alpha), \quad \text{Re}\{s\} < -\alpha$$

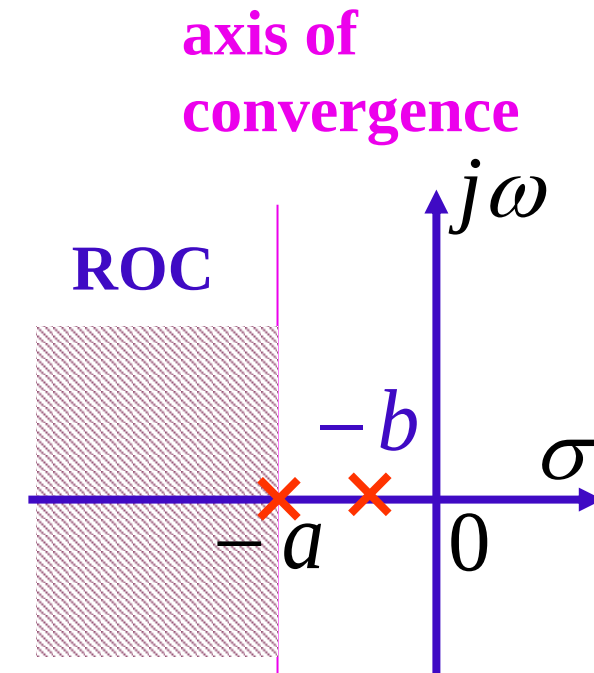
The ROCs of the bilateral Laplace transforms of the above two signals are both half-open planes on the s-plane. $-\alpha$ is called their abscissa of convergence, and $\sigma = \text{Re}\{s\} = -\alpha$ is called their **axis of convergence** or **boundary of the ROC**.

The same bilateral Laplace transform expression can represent different time signals due to different ROCs. Therefore, when representing the original signal by its bilateral Laplace transform, **the ROC must be specified; otherwise, the original signal cannot be determined.**

4.12 Bilateral Laplace Transform

Given the signal: $f(t) = (e^{-at} + e^{-bt})u(-t) \quad (a > b > 0)$

The image function is: $F(s) = \frac{-1}{s+a} + \frac{-1}{s+b} \quad (\sigma < -a)$



The ROC of a left-sided signal lies to the left of its leftmost pole.

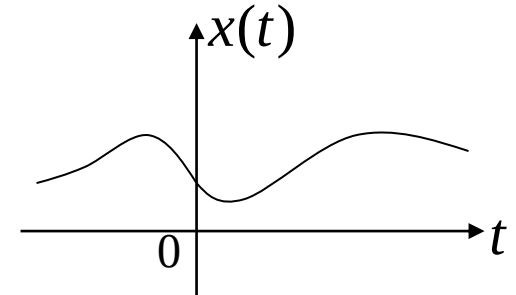
All poles of the image function are located to the right of the ROC.

4.12 Bilateral Laplace Transform

4.12.4 Bilateral Laplace Transform of Bilateral Signals

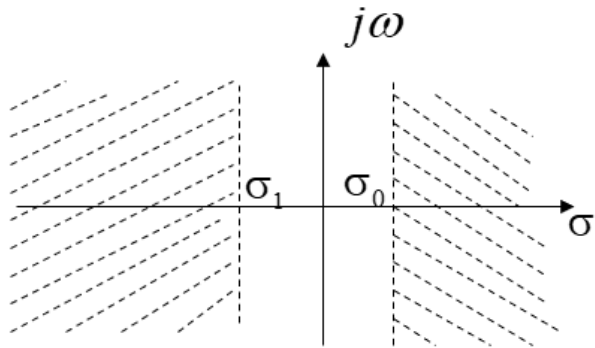
If a signal is a **bilateral signal**, its bilateral Laplace transform is:

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st} dt = \int_{-\infty}^0 x(t)e^{-\sigma t} e^{-j\omega t} dt + \int_0^{\infty} x(t)e^{-\sigma t} e^{-j\omega t} dt$$

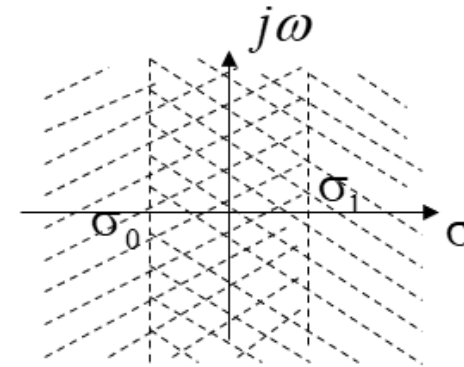


If σ_1 makes the first term of the above expression absolutely integrable, and σ_0 makes the second term absolutely integrable, then:

$$X(s) = \int_{-\infty}^0 x(t)e^{-\sigma_1 t} e^{-j\omega t} dt + \int_0^{\infty} x(t)e^{-\sigma_0 t} e^{-j\omega t} dt$$



if $\sigma_0 > \sigma_1$, $X(s)$ does not exist



if $\sigma_0 < \sigma_1$, $X(s)$ exists,
the ROC is $\sigma_0 < \sigma < \sigma_1$

Given $f(t) = u(t) + e^t u(-t)$, its bilateral Laplace Transform is ()

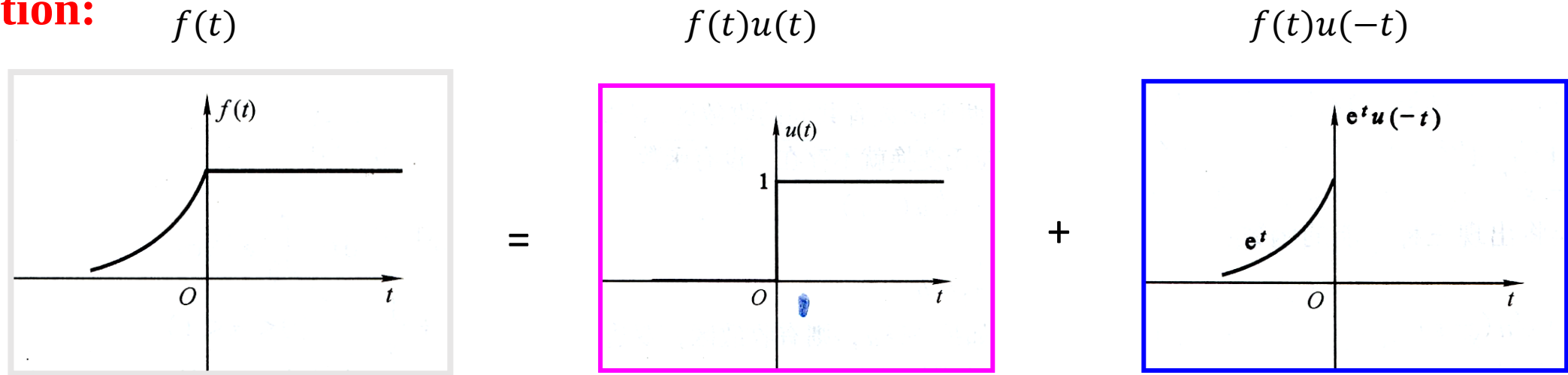
- A $F(s) = \frac{1}{s} - \frac{1}{s-1}$, $\sigma > 0$
- B $F(s) = \frac{1}{s} + \frac{1}{s+1}$, $0 < \sigma < 1$
- ☒ C $F(s) = \frac{1}{s} - \frac{1}{s-1}$, $0 < \sigma < 1$
- D $F(s) = \frac{1}{s} + \frac{1}{s+1}$, $\sigma > -1$

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4.12 Bilateral Laplace Transform

Example 4-33: Given $f(t) = u(t) + e^t u(-t)$, find its bilateral Laplace Transform.

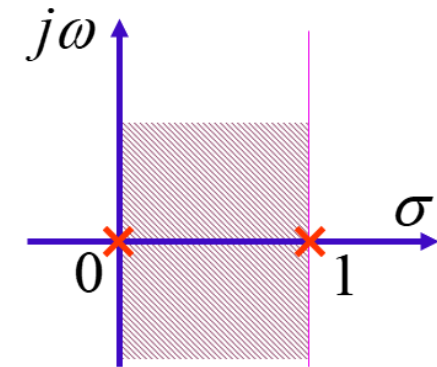
Solution:



$$F(s) = \int_{-\infty}^{\infty} f(t) e^{-st} dt = \int_{-\infty}^0 e^t e^{-st} dt + \int_0^{\infty} e^{-st} dt = \int_{-\infty}^0 e^{(1-\sigma)t} e^{-j\omega t} dt + \int_0^{\infty} e^{-\sigma t} e^{-j\omega t} dt$$

To make the two terms in the above expression absolutely integrable, $1 - \sigma > 0$ and $\sigma > 0$, so $0 < \sigma < 1$.

$$F(s) = \frac{1}{s} + \frac{1}{1-s} \quad (0 < \sigma < 1)$$



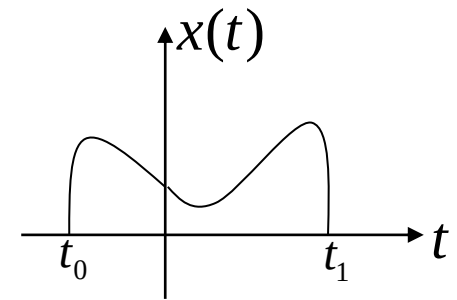
ROC

4.12 Bilateral Laplace Transform

4.12.5 Bilateral Laplace Transform of Time-Limited Signals

If a signal is a **time-limited signal**, i.e. $t_0 < t < t_1$, $x(t)=0$, its Laplace transform is defined as follows:

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st} dt = \int_{t_0}^{t_1} x(t)e^{-st} dt = \int_{t_0}^{t_1} x(t)e^{-\sigma t} e^{-j\omega t} dt$$

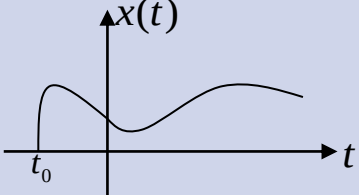
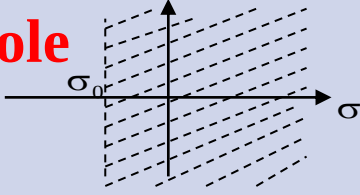
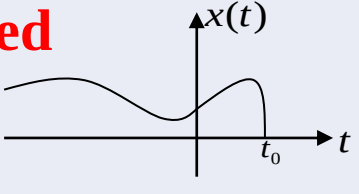
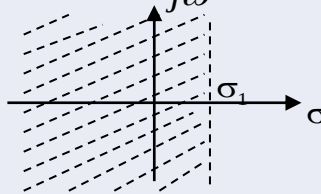
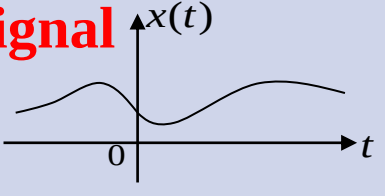
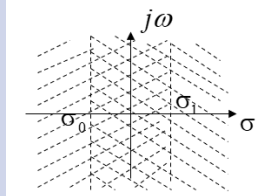
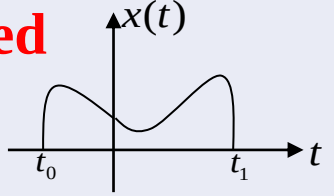
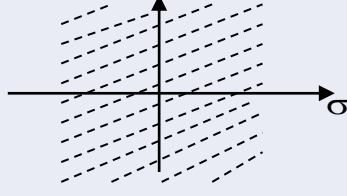


At this time, if $x(t)$ is absolutely integrable, $x(t)e^{-\sigma t}$ is always absolutely integrable. Therefore, its bilateral Laplace transform exists over the entire s-plane, **i.e. the ROC is the entire s-plane.**

4.12 Bilateral Laplace Transform

4.12.6 Characteristics of the ROC for Bilateral Laplace Transform

When considering the existence of the ROC for the bilateral Laplace transform, **the ROC must be specified.**

Signal Category	ROC of Bilateral Laplace Transform
Right-Sided Signal 	To the right of the rightmost pole of the image function 
Left-Sided Signal 	To the left of the leftmost pole of the image function 
Bilateral Signal 	Band-shaped region between two poles (if the bilateral Laplace transform exists) 
Time-Limited Signal 	Entire s-plane 

4.12 Bilateral Laplace Transform

4.12.7 Inverse Bilateral Laplace Transform

It can be solved based on the ROC and the poles:

- (1) Perform partial fraction decomposition on the image function;
- (2) Take the inverse transform of the image function according to the poles;
- (3) **Poles to the left of the ROC correspond to right-sided signals, and poles to the right of the ROC correspond to left-sided signals.**

F(s)	ROC	f(t)
$1/s$	$\sigma > 0$	$u(t)$
$1/s$	$\sigma < 0$	$-u(-t)$
$1/(s+\alpha)$	$\sigma > -\alpha$	$e^{-\alpha t} u(t)$
$1/(s+\alpha)$	$\sigma < -\alpha$	$-e^{-\alpha t} u(-t)$

4.12 Bilateral Laplace Transform

Example 4-34: Given $F(s) = \frac{1}{s} + \frac{1}{s + \alpha}$ ($\alpha > 0$), find all possible inverse bilateral Laplace transforms.

Solution:

According to the poles of the image function, 0 and $-\alpha$, there are **three possible cases** for its ROC, and different ROCs yield different time-domain signals when taking the inverse transform.

$\sigma > 0$	$f(t) = u(t) + e^{-\alpha t}u(t)$
$-\alpha < \sigma < 0$	$f(t) = -u(-t) + e^{-\alpha t}u(t)$
$\sigma < -\alpha$	$f(t) = -u(-t) - e^{-\alpha t}u(-t)$

4.13 Relationship between Laplace Transform and Fourier Transform

The Laplace transform is an extension of the Fourier transform from the real frequency ω to the complex frequency $s = \sigma + j\omega$, and the Fourier transform is a special case of the Laplace transform on the imaginary axis of the s -plane.

$$X(s) = \mathcal{L}\{x(t)\} = \mathcal{F}\{x(t)e^{-\sigma t}\} = X(\sigma + j\omega)$$

For many signals, replacing $j\omega$ with s in their Fourier transform gives their Laplace transform, and vice versa. For example, the one-sided exponential decay signal:

$$e^{-\alpha t}u(t) \xleftrightarrow{FT} \frac{1}{j\omega + \alpha} \quad \therefore e^{-\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{s + \alpha}$$

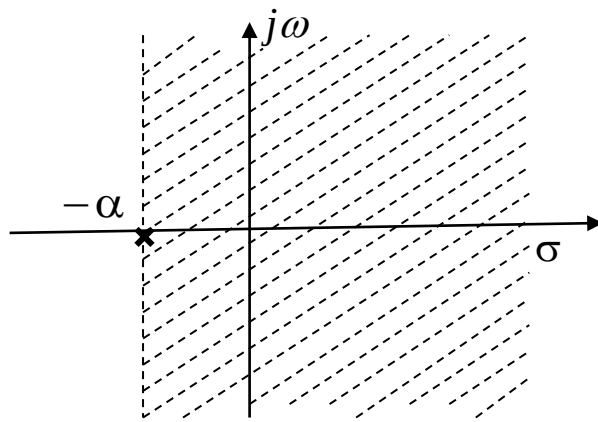
However, there are also **exceptions**. For example, the unit step signal:

$$u(t) \xleftrightarrow{FT} \pi\delta(\omega) + \frac{1}{j\omega} \quad u(t) \xleftrightarrow{LT} \frac{1}{s}$$

4.13 Relationship between Laplace Transform and Fourier Transform

If the ROC of a signal's Laplace transform includes the imaginary axis, its Fourier transform can be obtained by substituting $s=j\omega$ into the Laplace transform.

For example, the poles of a one-sided exponential decay signal lie on the negative real axis.



ROC of Laplace Transform for $e^{-\alpha t}u(t)$

Example of a multiple pole on the negative real axis:

$$te^{-\alpha t}u(t) \xleftrightarrow{FT} \frac{1}{(j\omega + \alpha)^2}$$

$$te^{-\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{(s + \alpha)^2}$$

Example of conjugate complex poles with negative real:

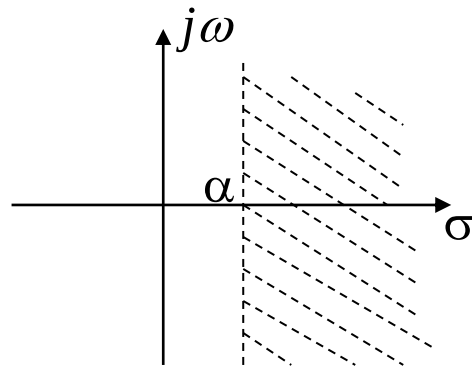
$$e^{-\alpha t} \cos(\Omega_0 t)u(t) \xleftrightarrow{FT} \frac{j\omega + \alpha}{(j\omega + \alpha)^2 + \Omega_0^2}$$

$$e^{-\alpha t} \cos(\Omega_0 t)u(t) \xleftrightarrow{LT} \frac{s + \alpha}{(s + \alpha)^2 + \Omega_0^2}$$

4.13 Relationship between Laplace Transform and Fourier Transform

If the ROC of the Laplace transform does not include the imaginary axis, i.e., the poles of the signal's Laplace transform lie on the right-half plane of the s-plane, its Fourier transform does not exist.

For example, a one-sided exponentially growing signal has its poles on the positive real axis.



$$e^{\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{s - \alpha}$$

ROC of Laplace Transform for $e^{\alpha t}u(t)$

Since the signal grows exponentially and does not satisfy the condition of absolute integrability, its Fourier transform does not exist.

4.13 Relationship between Laplace Transform and Fourier Transform

When the Laplace transform of a signal **has poles located on the imaginary axis of the s-plane, we cannot simply substitute $j\omega$ for s to obtain its Fourier transform.** For example, the unit step signal $u(t)$:

$$u(t) \xleftrightarrow{LT} \frac{1}{s} \qquad u(t) \xleftrightarrow{FT} \pi\delta(\omega) + \frac{1}{j\omega}$$

Suppose the Laplace transform $X(s)$ of a signal $x(t)$ has a k -order pole on the imaginary axis: $j\Omega_0$

$$X(s) = \underset{\nearrow}{X_1(s)} + \frac{A_0}{(s - j\Omega_0)^k}$$

Poles are located to the left of the imaginary axis

It can be proved that its corresponding Fourier transform is:

$$X(\omega) = X(s) \big|_{s=j\omega} + \frac{\pi A_0 j^{k-1}}{(k-1)!} \delta^{(k-1)}(\omega - \Omega_0)$$

- Repeat the examples in the lecture notes.
- 6.43(a) and 6.50(b) in the textbook.